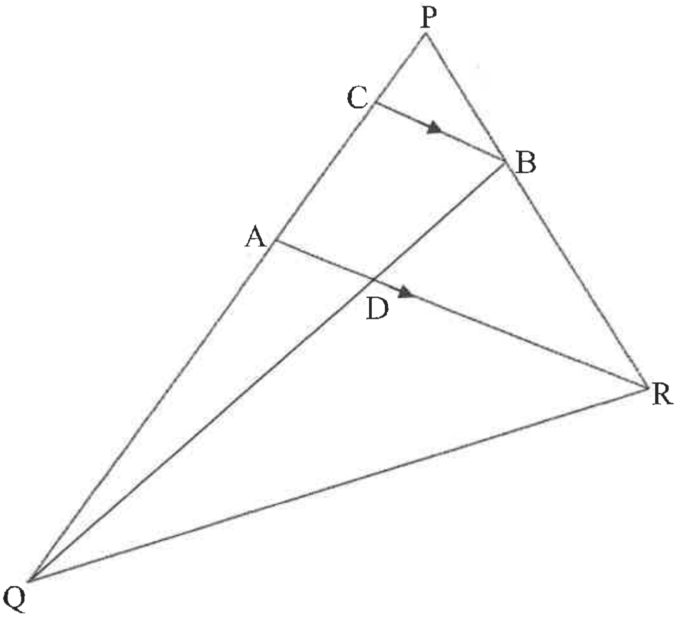
12 + 13 # Euclidean Geometry 2

* Remember the proofs of the theorems

QUESTION 1

In ΔPQR below, B lies on PR such that $2PB = BR$. A lies on PQ such that [redacted] $PA : PQ = 5 : 9$
BC is drawn parallel to AR.

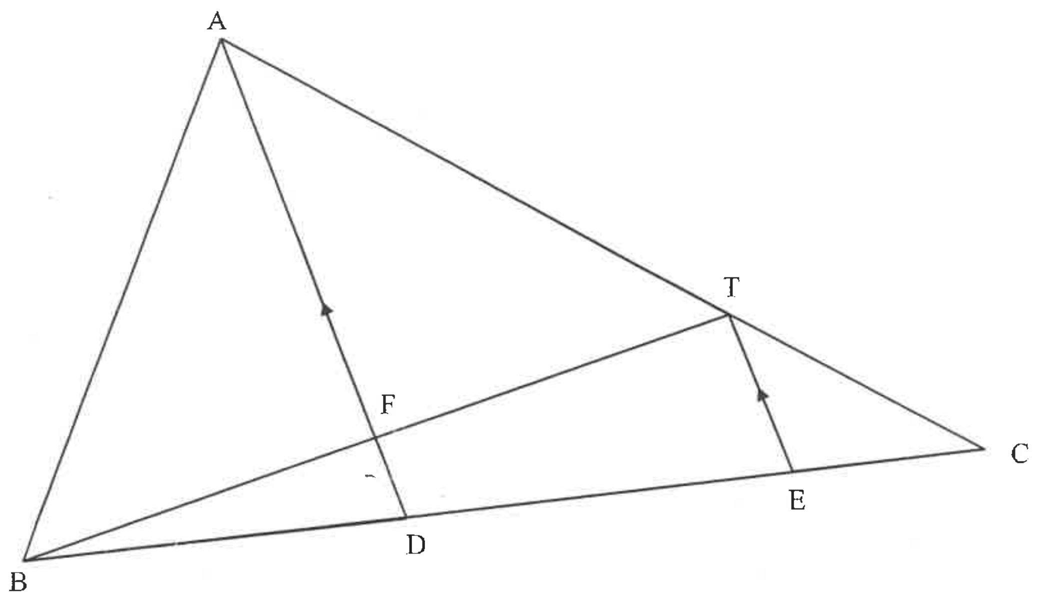


1.1 Write down the value of $\dfrac{\text{area of } \Delta PRA}{\text{area of } \Delta QRA}$ (2)

1.2 Calculate the value of the ratio $\dfrac{BD}{BQ}$. Show all working to support your answer. (5)

QUESTION 2

In the figure below, ΔABC has D and E on BC. BD = 6 cm and DC = 9 cm.
AT : TC = 2 : 1 and AD || TE.



2.1 Write down the numerical value of $\dfrac{CE}{ED}$ (1)

2.2 Show that D is the midpoint of BE. (2)

2.3 If FD = 2 cm, calculate the length of TE. (2)

2.4 Calculate the numerical value of:

2.4.1 $\dfrac{\text{Area of } \Delta ADC}{\text{Area of } \Delta ABD}$ (1)

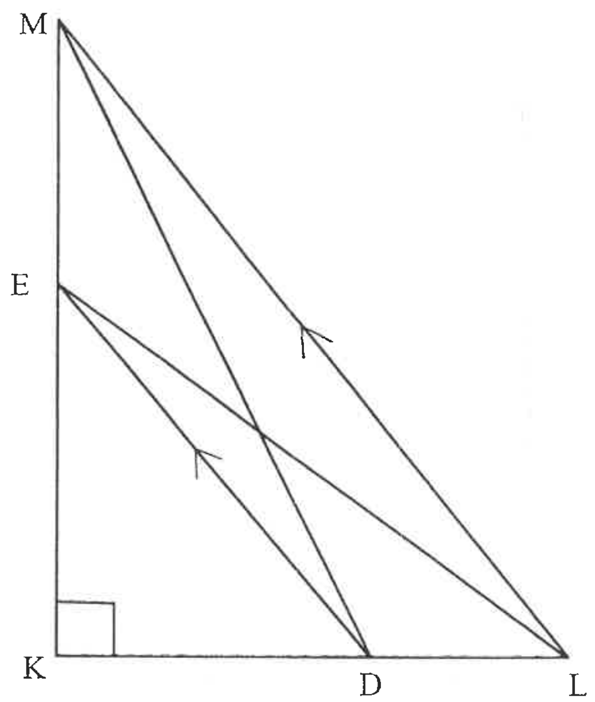
2.4.2 $\dfrac{\text{Area of } \Delta TEC}{\text{Area of } \Delta ABC}$ (3)

QUESTION 3

3.1 Complete the following statement:

A line parallel to one side of a triangle divides … (1)

3.2 In Δ KLM, $\hat{K} = 90^\circ$ and D is a point on KL so that KD:DL = 2:1 and DE // LM with E on KM.

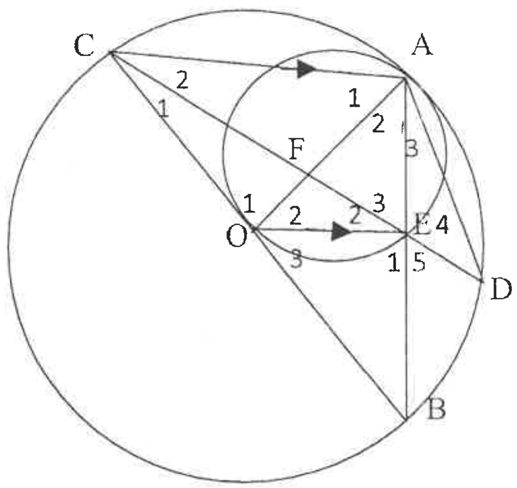


3.2.1 If DL = x and EM = y, express LM^2 in terms of x and y. (4)

3.2.2 Show that $DM^2 + LE^2 = \frac{13}{9}LM^2$. (4)

QUESTION 4 :

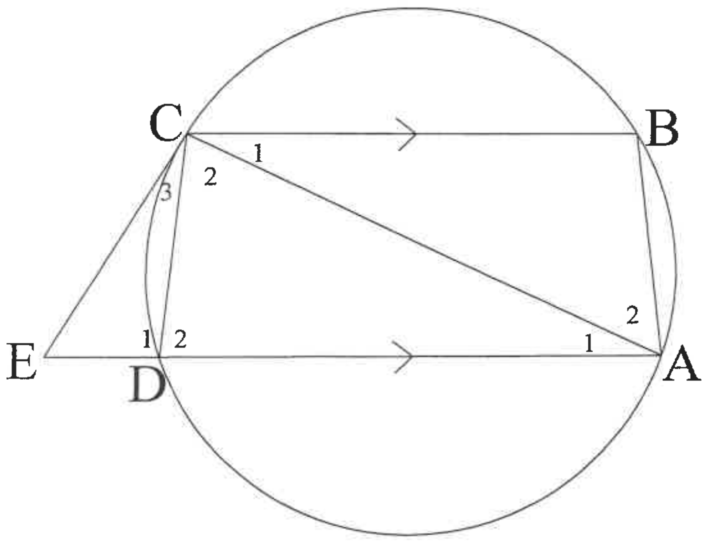
Two circles touch each other internally at A. The smaller circle passes through O, which is the centre of the larger circle. E lies on the circumference of the smaller circle. A, D, B and C are points on the circumference of the larger circle. OE || CA.



4.1	Prove, with reasons, that AE = EB.	(3)
4.2	Prove that Δ AED ||| ΔCEB.	(3)
4.3	Hence, or otherwise, show that $AE^2 = DE.CE$.	(3)
4.4	If $AE.EB = EF.EC$, show that E is the midpoint of DF.	(3)

QUESTION 5

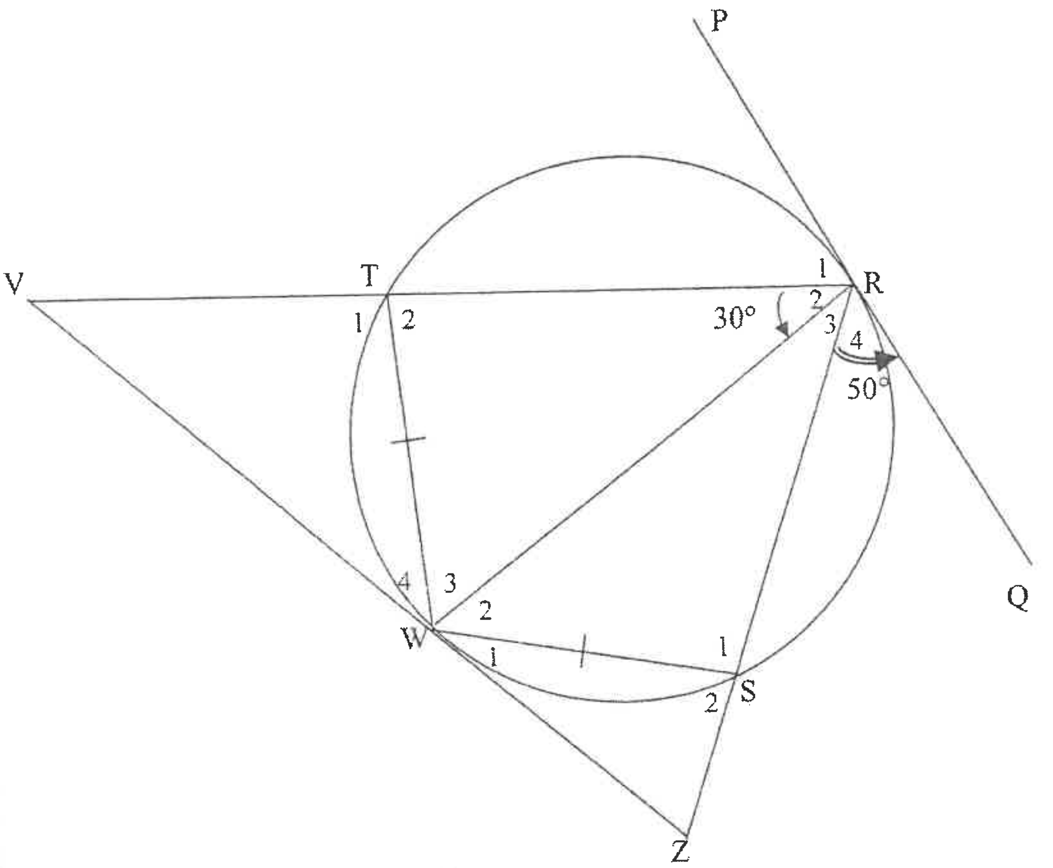
In the following diagram, CE is a tangent to circle ABCD at C. CB || EA.



Prove that $AB \cdot CD = BC \cdot ED$. (7)

QUESTION 6

In the figure, TRSW is a cyclic quadrilateral with TW = WS. RT and RS are produced to meet tangent VWZ at V and Z respectively. PRQ is a tangent to the circle at R. RW is joined. $\hat{R}_2 = 30°$ and $\hat{R}_4 = 50°$.



6.1 Give a reason why $\hat{R}_3 = 30°$. (1)

6.2 State, with reasons, TWO other angles equal to 30°. (3)

6.3 Determine, with reasons, the size of:

(a) $\hat{S}_2$ (3)

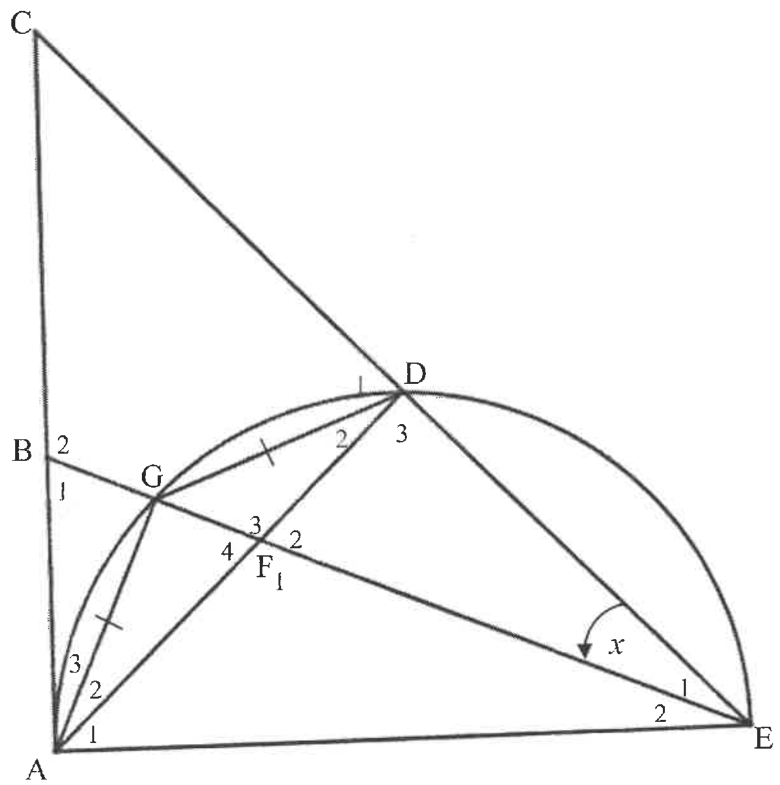
(b) $\hat{V}$ (4)

6.4 Prove that $WR^2 = RV \times RS$. (5)

QUESTION 7

In the figure AGDE is a semicircle. AC is the tangent to the semicircle at A and EG produced intersects AC at B. AD intersects BE in F.

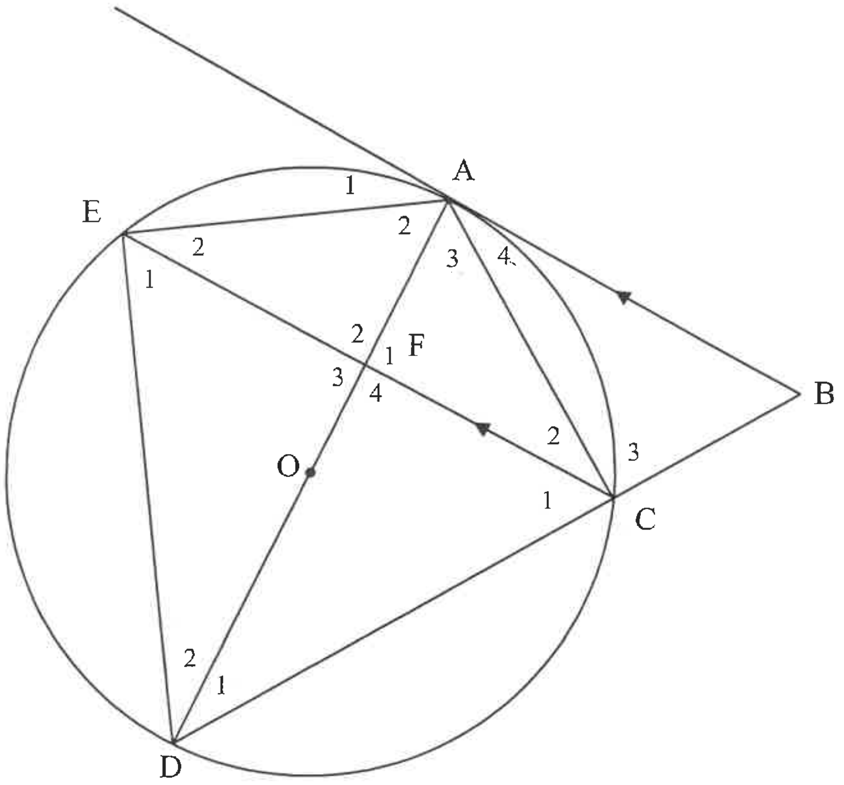
AG = GD. $\hat{E}_1 = x$.



7.1	Write down, with reasons, FOUR other angles each equal to x.	(8)
7.2	Prove that BE.DE = AE.FE	(7)
7.3	Prove that $\hat{B}_1 = \hat{D}_1$	(4)

QUESTION 8

In the figure below, AB is a tangent to the circle with centre O. AC = AO and BA || CE. DC produced, cuts tangent BA at B.



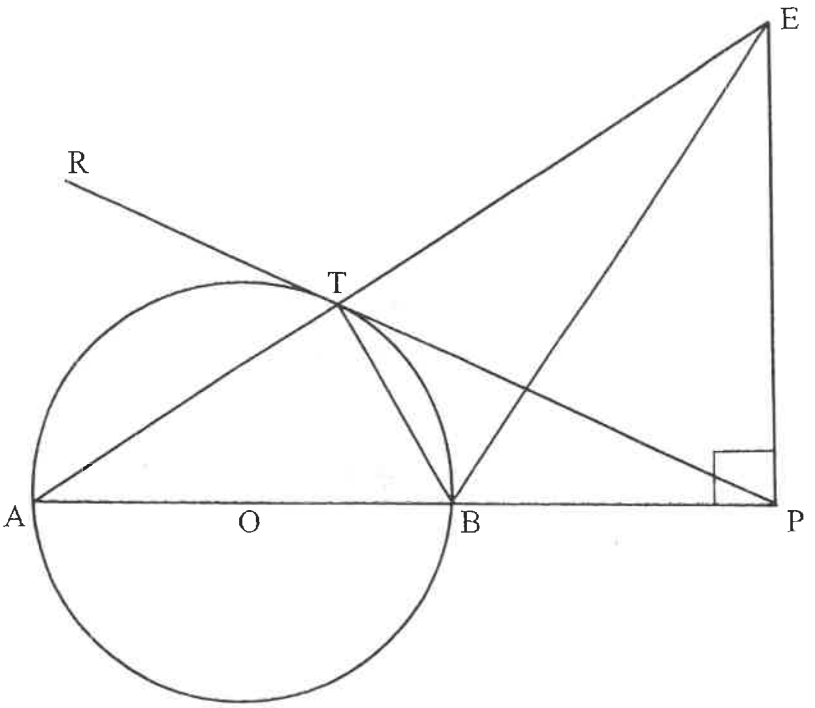
8.1 Show $\hat{C}_2 = \hat{D}_1$. (3)

8.2 Prove that $\Delta ACF \mathbin{|||} \Delta ADC$. (3)

8.3 Prove that AD = 4AF. (4)

QUESTION 9

In the diagram below, O is the centre with A, B and T on the circumference, $BP = OB = AO$, PTR is a tangent and $EP \perp AP$.



Prove that:

9.1	TEPB is a cyclic quad	(3)
9.2	$\Delta ATB \ /// \ \Delta APE$	(3)
9.3	$TP = PE$	(6)
9.4	$\Delta ATB \ /// \ \Delta EPB$	(5)
9.5	$2BP^2 = BE.TB$	(4)

Euclidean Geometry 2: Solutions

Question 1:

1.1. $\dfrac{\text{area } \Delta PRA}{\text{area } \Delta QRA}$

$= \dfrac{PA}{QA}$

$= 5/4$

1.2. $\dfrac{AC}{AP} = \dfrac{BR}{PR}$ line // one side Δ

$\dfrac{AC}{5m} = \dfrac{2k}{3k}$

$AC = \dfrac{10}{3} m$

$\dfrac{BD}{DQ} = \dfrac{AC}{AQ}$ line // one side Δ

$= \dfrac{10m}{3} \div 9m$

$= \dfrac{10m}{3} \times \dfrac{1}{9m}$

$= \dfrac{10}{27}$

Question 2:

2.1. $\frac{CE}{ED} = \frac{CT}{AT}$ line ∥ one side Δ

$= \frac{k}{2k}$

$= \frac{1}{2}$

2.2. $\frac{DE}{DC} = \frac{AT}{AC}$ line ∥ one side Δ

$\frac{DE}{9} = \frac{2k}{3k}$

$DE = 6\text{ cm}$

$\therefore BD = DE$

2.3 $TE = 2(FD)$ mdpt theorem

$= 2(2)$

$= 4\text{ cm}$

2.4.1. $\frac{\text{Area } \Delta ADC}{\text{Area } \Delta ABD}$

$= \frac{DC}{BD}$

$= \frac{9}{6}$

$= \frac{3}{2}$

2.4.2. $\frac{\text{Area } \triangle\ TEC}{\text{Area } \triangle\ ABC}$

$= \frac{\frac{1}{2}\ TC \cdot EC \ \sin C}{\frac{1}{2}\ AC \cdot BC \ \sin C}$

$= \frac{TC \cdot EC}{AC \cdot BC}$

$= \frac{K \cdot 3}{3K \ 15}$

$= \frac{1}{15}$

Question 3

3.1. the other two sides in proportion

3.2. $KD = 2x$

$\therefore KL = 3x$

$\frac{MK}{EM} = \frac{KL}{DL}$ line // one side $\triangle$

$\frac{MK}{y} = \frac{3x}{x}$

$MK = 3y$

$MK^2 + KL^2 = LM^2$ pythag

$(3y)^2 + (3x)^2 = LM^2$

$9x^2 + 9y^2 = LM^2$

3.3. $EK = 2y$

In $\triangle$ DKM :

$DM^2 = KM^2 + DK^2$ pythag
$= (3y)^2 + (2x)^2$
$= 9y^2 + 4x^2$

In $\triangle$ EKL :

$EL^2 = KL^2 + EK^2$ pythag
$= (3x)^2 + (2y)^2$
$= 9x^2 + 4y^2$

$DM^2 + EL^2$
$9y^2 + 4x^2 + 9x^2 + 4y^2$
$= 13x^2 + 13y^2$
$= 13(x^2 + y^2)$
$= \frac{13}{9}[9(x^2 + y^2)]$
$= \frac{13}{9} LM^2$

Question 4:

4.1. $OB = OC$ radii
$\frac{OB}{OC} = \frac{BE}{EA}$ line // one side $\triangle$
$\therefore \frac{BE}{EA} = 1$
$\therefore BE = EA$

4.2 In $\triangle AED$ and $\triangle CEB$

1) $\hat{A}_3 = \hat{C}_1$ $\angle$'s in same segment
2) $\hat{E}_4 = \hat{E}_1$ vert opp $\angle$'s
3) $\hat{D} = \hat{B}$ sum of $\angle$'s of $\triangle$

$\therefore \triangle AED \mathbin{|||} \triangle CEB$ ($\angle\angle\angle$)

4.3 $\dfrac{AE}{CE} = \dfrac{ED}{EB} = \dfrac{AD}{CB}$ $|||$ $\triangle$'s

but $BE = AE$ proven

$\therefore \dfrac{AE}{CE} = \dfrac{ED}{AE}$

$\therefore \underline{AE^2 = ED.CE}$

4.4 $AE . EB = EF . EC$ given
but $BE = AE$ proven

$\therefore AE^2 = EF . EC$

$\therefore ED.CE = EF.EC$ (both $= AE^2$)
$\therefore \underline{ED = EF}$

Question 5

In $\triangle$ BCA and $\triangle$ DCE

1) $\hat{B} = \hat{D}_1$ ext $\angle$ of cyclic quad.
2) $\hat{C}_1 = \hat{A}_1$ alt $\angle$'s BC // AE
$\hat{A}_1 = \hat{C}_3$ tan chord theorem
$\therefore \hat{C}_1 = \hat{C}_3$
3) $\hat{A}_2 = \hat{E}$ $\angle$'s of $\triangle$

$\therefore \triangle BCA \parallel\!\mid \triangle DCE$ ($\angle\angle\angle$)

$\therefore \dfrac{BC}{DC} = \dfrac{CA}{CE} = \dfrac{BA}{DE}$ ||| $\triangle$'s

$\therefore AB.CD = BC.ED$

Question 6

6.1. $\hat{R}_3 = \hat{R}_2 = 30°$ equal chords; equal $\angle$s

6.2. $\hat{W}_1 = 30°$ tan chord theorem
$\hat{W}_4 = 30°$ tan chord theorem

6.3. a) $\hat{W}_2 = 50°$ tan chord theorem
$\hat{S}_2 = \hat{W}_2 + \hat{R}_3$ ext $\angle$ of $\triangle$
$= \underline{80°}$

b) $\hat{T}_2 = \hat{S}_2 = 80°$ ext $\angle$ of cyclic quad
$\hat{T}_2 = \hat{V} + \hat{W}_4$ ext. $\angle$ of $\triangle$
$\hat{V} = 50°$

6.4 In $\triangle$ WRS and $\triangle$ VRW

1) $\hat{W}_2 = \hat{R}_4 = 50°$ tan chord theorem

$\hat{V} = 50°$ proven

$\therefore \hat{W}_2 = \hat{V}$

2) $\hat{R}_3 = \hat{R}_2$ proven

3) $\hat{S}_1 = \hat{W}_3 + \hat{W}_4$ sum of $\angle$'s of $\triangle$

$\therefore \triangle WRS \; ||| \; \triangle VRW$ $(\angle\angle\angle)$

$\therefore \frac{WR}{VR} = \frac{RS}{RW} = \frac{WS}{VW}$ ||| $\triangle$'s

$\therefore WR^2 = VR \cdot RS$

<u>Question 7</u>

7.1 $\hat{E}_2 = \hat{E}_1 = x$ equal chords; equal $\angle$'s

$\hat{A}_3 = \hat{E}_2 = x$ tan chord theorem

$\hat{D}_2 = \hat{E}_2 = x$ $\angle$'s in same segment

$\hat{A}_2 = \hat{E}_1 = x$ $\angle$'s in same segment

7.2. In $\triangle$ AEB and $\triangle$ DEF

1) $\hat{E}_2 = \hat{E}_1 = x$ proven

2) $\hat{A} = 90°$ tan $\perp$ radius

$\hat{D}_3 = 90°$ $\angle$ in semi circle

$\therefore \hat{A} = \hat{D}_3$

3) $\hat{B}_1 = \hat{F}_2$ sum of $\angle$'s of $\triangle$

$\therefore \triangle AEB \; ||| \; \triangle DEF$ $(\angle\angle\angle)$

$\frac{AE}{DE} = \frac{EB}{EF} = \frac{AB}{DF}$ ||| Δ's

$\therefore BE.DE = AE.EF$

7.3. $\hat{B}_1 = \hat{F}_2$ proven

$\hat{D}_1 + \hat{D}_2 = \hat{F}_2 + \hat{E}_1$ ext $\angle$ of Δ

$\hat{D}_1 + x = \hat{B}_1 + x$

$\therefore \underline{\hat{D}_1 = \hat{B}_1}$

Question 8

8.1. $\hat{C}_2 = \hat{A}_4$ alt $\angle$'s AB // CE

$\hat{A}_4 = \hat{D}_1$ tan chord theorem

$\therefore \hat{C}_2 = \hat{D}_1$ (both $= \hat{A}_4$)

8.2. In Δ ACF and Δ ADC

1) $\hat{A}_3 = \hat{A}_3$ common

2) $\hat{C}_2 = \hat{D}_1$ proven

3) $\hat{F}_1 = \hat{C}_1 + \hat{C}_2$ sum of $\angle$'s of Δ

$\therefore \Delta$ ACF ||| Δ ADC ($\angle\angle\angle$)

8.3 $\frac{AC}{AD} = \frac{CF}{DE} = \frac{AF}{AC}$ ||| Δ's

$\therefore \frac{AC}{AD} = \frac{AF}{AC}$

$\frac{AO}{AD} = \frac{AF}{AO}$ (AO = AC given)

$\frac{\frac{1}{2}AD}{AD} = \frac{AF}{\frac{1}{2}AD}$ (AO $= \frac{1}{2}$ AD)

$$\frac{1}{2} = \frac{2\,AF}{AD}$$

$$AD = 4\,AF$$

Question 9

9.1. $A\hat{T}B = 90^\circ$ $\quad$ ∠ in a semi circle

$\hat{P} = 90^\circ$

$\therefore\ A\hat{T}B = \hat{P}$

$\therefore$ TEPB is a cyclic quad

ext ∠ of quad = opp int ∠.

9.2. In $\triangle$ ATB and $\triangle$ APE

1) $\hat{A} = \hat{A}$ $\quad$ common

2) $A\hat{T}B = \hat{P}$ $\quad$ proven

3) $A\hat{B}T = A\hat{E}P$ $\quad$ sum of ∠'s of $\triangle$

$\therefore\ \triangle ATB\ |||\ \triangle APE$ $\quad$ (∠∠∠)

9.3. $B\hat{T}E = 90^\circ$ $\quad$ ∠'s on a straight line

$P\hat{T}E = 90^\circ - B\hat{T}P$

$\hat{A} = B\hat{T}P$ $\quad$ tan chord theorem

$A\hat{B}T = 180^\circ - 90^\circ - \hat{A}$ $\quad$ int ∠'s of $\triangle$

$= 90^\circ - \hat{A}$

$= 90^\circ - B\hat{T}P$

$A\hat{B}T = \hat{E} = 90 - B\hat{T}P$ $\quad$ proven

$\therefore\ P\hat{T}E = \hat{E}$

$\therefore\ PT = PE$ $\quad$ sides opp equal ∠'s

9.4 In $\triangle$ ATB and $\triangle$ EPB

1) $P\hat{E}B = B\hat{T}P$ $\angle$'s in same segment
$B\hat{T}P = \hat{A}$ tan chord theorem
$\therefore \hat{A} = P\hat{E}B$

2) $A\hat{T}B = \hat{P} = 90°$ proven

3) $A\hat{B}T = E\hat{B}P$ sum of $\angle$'s of $\triangle$

$\therefore \triangle ATB \parallel\parallel \triangle EPB$ ($\angle\angle\angle$)

9.5 $\frac{AT}{EP} = \frac{TB}{PB} = \frac{AB}{EB}$ $\parallel\parallel$ $\triangle$'s

but $BP = AO = OB$ given
$\therefore AB = 2BP$

$\therefore \frac{2BP}{BE} = \frac{TB}{BP}$
$2BP^2 = BE \cdot TB$